American International University-Bangladesh (AIUB)

Department of Computer Science Faculty of Science & Technology (FST)

PROJECT TITLE

Online Banking System : A Full-Stack Digital Finance Platform

A Software Engineering Project Submitted
By

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SL	**Student Name**	**Student ID**
1	Md. Mohaimanul Haque	24-56377-1
2	Sheikh MD Sabbir Hossain	23-54934-3
3	Abid Arman Saif	23-55624-3
4	Yasir Zaman Tasfin	23-52411-2
5	Md Sifat Hassan	24-56372-1

Online Banking System: A Full- Stack Digital Finance Platform

1. Project Title

Online Banking System: A Full-Stack Digital Finance Platform for Account Management, Fund Transfers, and Secure Transaction Processing.

2. Introduction

2.1 Overview of the Domain

The digital banking sector has redefined personal finance, with millions of users shifting from physical branch visits to instant mobile and web-based transactions. Modern financial platforms integrate real-time balance updates, secure multi-factor authentication, and personalized financial insights. Utilizing a robust MERN stack allows for the creation of scalable, high-performance banking solutions that provide users with 24/7 access to their funds. This project focuses on a Retail Banking model, allowing individual customers to manage their accounts directly through a secure web storefront.

2.2 Motivation of the Project

Building a full-stack banking system is the perfect way to practice core Software Engineering principles—like system design, testing, and deployment—in a high-stakes, real-world scenario. Many current banking interfaces are clunky or expensive for smaller financial institutions to maintain. This project aims to show that we can build a secure, custom alternative within a single semester while creating a flagship portfolio piece that proves our full-stack skills.

3. Background to the Problem

3.1 Current System Overview

Many people still deal with outdated banking methods, such as visiting physical branches for simple tasks, using manual paper-based tracking, or relying on basic spreadsheets to monitor their spending. These old-school approaches are slow, disorganized, and lack the security needed for modern finance.

3.2 Limitations of the Existing System

Key limitations include: (a) Lack of Accessibility: No centralized, 24/7 searchable transaction history. (b) Inefficient Processing: Manual handling of fund transfers leading to delays and human error. (c) Security Risks: Insecure storage of sensitive user data and no digital audit trails. (d) Limited Insights: No automated financial analytics or spending reports for the user. (e) Poor Scalability: Difficulty managing an increasing volume of concurrent transactions.

3.3 Need for the Proposed Solution

An online banking platform fixes these issues by offering a structured dashboard, secure transfers powered by modern APIs, and real-time status tracking for every transaction. It gives the institution full control over its data without the high costs of third-party platforms

4. Problem Statement

Customers of smaller financial institutions lack an integrated, secure digital platform for daily banking needs. Without a centralized system, they cannot perform instant transfers, view persistent transaction histories, or manage accounts remotely. This leads to operational friction, high costs for physical processing, and security risks from manual data handling. A custom-built, full-stack banking platform is required to provide end-to-end financial capabilities with complete data ownership.

5. Objectives of the Project

1. Develop a responsive account dashboard with balance overviews and transaction history.

2. Implement a secure internal fund transfer system with real-time balance validation.

3. Integrate robust security protocols, including encrypted data handling and transaction logging.

4. Build a statement generation module with filters for date, amount, and transaction type.

5. Design a role-based admin dashboard for account approval, fraud monitoring, and system analytics.

6. Implement JWT-based authentication to keep user data private and secure.

6. Solution Overview

We propose a full-stack MERN application (MongoDB, Express.js, React.js, Node.js) built on a three-tier architecture. This includes a React frontend for a smooth user experience, a RESTful API backend, and a MongoDB database for flexible storage.

Technology Stack

Component	Technology	Rationate
Frontend	React.js + Redux	ComponenL-based UI with centralized state for account balances.
Backend	Node.js / Express.js	Non-blocking I/O for handling high frequency transaction requests.
Database	MongoDB / Mongoose	Flexible schema for diverse transaction types and user profiles.
Authentication	JWT + bcrypt	Stateless token auth with secure password hashing for financial data.
Security	SSL/TLS + Validation	Ensuring data integrity and secure transmission during transfers.
Cloud Storage	Cloudinary/AWS	Secure hosting for user KYC documents and digital statements.

7. Target Users

7.1 Primary Users

Customers: View balances, transfer funds, manage profiles, and download statements. **Bank Administrators:** Manage user accounts, monitor for suspicious activity, and approve transfer.

7.2 Secondary Users

Financial Auditors: Access transaction logs and reports for regulatory compliance. **Customer Support:** Assist users with account lockouts or transaction inquiries.

8. Scope of the Project

8.1 In-Scope

Secure user login and role-based access; Account dashboard with search and transaction filters; Secure fund transfers and order confirmations; Admin tools for account CRUD, transaction monitoring and analytics; Fully responsive design deployed on the cloud.

8.2 Out-of-Scope

Multi-currency support or ML-based financial advice; Live chatbot support or loyalty reward programs; Loan approval AI or credit scoring engines; Cryptocurrency wallets or trading features; External SWIFT/International wire transfers.

9. Expected Impact and Benefits

Operational: 70-80% reduction in manual data entry and branch paperwork. **Customer Experience:** 24/7 banking availability with transparent, self-service tracking. **Business Intelligence:** Analytics dashboard providing trends on deposits and withdrawals. **Academic:** Comprehensive SE lifecycle practice demonstrating full-stack proficiency.

10. Conclusion

This proposal presents a full-stack Online Banking System addressing the need for secure, digital-first financial management. Built on the MERN stack, it replaces fragmented, manual processes with a centralized, automated platform. The scope is realistic for a single-semester timeline, ensuring a functional and secure deliverable.